

Antioxidant effects of *Artemis sphaerocephala* Krasch. gum, on streptozotocin-induced type 2 diabetic rats

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ABSTRACT

Since ancient times, practitioners of traditional Chinese medicine have discovered that *Artemis sphaerocephala* Krasch. (Compositae) seed was useful for the treatment of diabetes. *A. sphaerocephala* Krasch. gum (ASK gum) is the water extract from the seed powder of the plant and has been used as a novel food additive by the food industry in China. The objective of this study was to evaluate the antioxidant function of ASK gum on type 2 diabetes using Streptozotocin-induced type 2 diabetic Sprague–Dawleys rats as an animal model; the antioxidant property of ASK gum was measured by the levels of superoxide dismutase (SOD), malondialdehyde (MDA), and $\cdot\text{OH}$, in liver and serum respectively. Normal rats and diabetic rats treated with metformin were used as the negative and positive control respectively. The results showed that a high-fat diet in combination with a low-dose Streptozotocin injection successfully induced type 2 diabetes among the experiment rats. The addition of ASK gum to the high-fat diet fed to the diabetic rats significantly increased their liver and serum SOD activities, however, the MDA and $\cdot\text{OH}$ levels in both liver and serum were significantly reduced. In addition, ASK gum markedly lessened the oxidative damage of kidney in the diabetic rats. A dose-dependent was observed between the measured biological parameters and the amount of ASK gum added. ASK gum at a level of 2.7% provided a similar protective effect to the diabetic rats as metformin.

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1. Introduction

Diabetes mellitus is one of the five leading causes of death worldwide. Abundant studies have demonstrated the presence of oxidative stress in type 2 diabetes mellitus (Ceriello & Motz, 2004; Guzik et al., 2002; Maritim, Sanders, & Watkins, 2003), a condition which exists in about 90% of diabetics (Damon, 1991; Mokuda, Okazaki, & Sakamoto, 2001; Yamamoto, Takakura, Kawamura, Seki, & Goto, 2001). Decreased concentration of anti-oxidative enzymes such as superoxide dismutase (SOD), and increased level of peroxidation products malondialdehyde (MDA) and reactive oxygen species (ROS) were detected in type 2 diabetes (Bhatia, Shukla, Venkata, Kaur, & Madhava, 2003; Elgebaly, Portik-Dobos, & Sachidanandam, 2007; Telci et al., 2000). ROS, including superoxide radical ($\text{O}_2^{\cdot-}$), alkoxyl radical ($\text{RO}^{\cdot}$), peroxy radical ($\text{ROO}^{\cdot}$), hydroxyl radical ($\cdot\text{OH}$) and hydrogen peroxide (H_2O_2), are unstable molecules. When the generation of ROS exceeds cellular defense

mechanisms, the ROS interact with biological macromolecules such as lipids, proteins and DNA, which could cause structural changes as well as functional abnormalities (Cheeseman & Slater, 1993). For this reason, oxidative stress has been considered to be a common pathogenic factor in the development of type 2 diabetes and its complications (Bhatia et al., 2003). Recently, there has been considerable interest in identifying effective agents from botanical sources, which may alleviate oxidative stress, thus assist in managing diabetes mellitus.

Artemis sphaerocephala Krasch. (ASK) is a low-growing Compositae shrub in the desert land of northwest China (Wei & Tu, 1980). The gums extracted from ASK seed have many desirable physical properties as a food ingredient for improving structure and quality of various foods (Hu, Yang, Du, & Zhang, 2006; Wei, Zhang, & Wang, 1996). Commercial ASK gum is available in China as a food additive (Hu et al., 2006).

According to the ancient anagraph of traditional Chinese medicine, ASK seeds have been used to alleviate many diseases including diabetes, parotitis, tonsillitis, scabies and ileuses (Zhao & Huang, 1981). However, the active compounds responsible for these therapeutic effects have not been clearly identified so far.

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There are a few recent patents which claimed that ASK gum had hypoglycemic and antioxidant effect on type 1 diabetic rats and hypolipidemic activity on obese mice (Berijhangeriletu et al., 2005; Hu & Xing, 2009; Su, Sun, Dong, Han, & Zhai, 2008; Zhang et al., 2005). In a previous study, we also found that ASK gum could alleviate hyperglycemia, hyperlipemia and insulin resistance in type 2 diabetes in a rat model (unpublished data), however, the mechanisms by which ASK gum exerts these physiological effects are not known. The objective of this study thus was to investigate the anti-oxidative function of ASK gum, which may be responsible for, or at least, contribute to the above mentioned physiological properties. A type 2 diabetic rat model similar to that reported previously (Reed et al., 2000) was used in this study. To the best of our knowledge, there is no report in the literature on the evaluation of antioxidant effect of ASK gum on type 2 diabetes using animal models.

2. Materials and methods

2.1. Plant material

A. sphaerocephala Krasch. seeds were harvested in 2007 in the sandy area of Yan'an, Shaanxi Province. After removing impurities, the seeds were dried in a warm air dryer at 38 °C for 24 h. After drying, the seeds were ground into powders with particle diameter <0.8 mm using a Perten 3303 Mill (Perten Instrument Company, Sweden).

2.2. Extraction of ASK gum

ASK powder was dispersed in water (1:20, w/w) at 25 °C for 2 h under constant stirring. The mixture was homogenized twice during the extraction, each time for 5 min, using a Polytron homogenizer (Kinematica GmbH/Brinkmann Instruments, Rexdale, ON, Canada) at speed 11. The suspension was then centrifuged at 9720 g for 15 min to remove residues. The extracted gum in the supernatant was precipitated by adding excess ethanol (1:3, v/v), followed by centrifugation at 1000 g for 5 min. The pellet was washed with 70% alcohol, left in 100% isopropyl alcohol at 4 °C for 12 h, then frozen – vacuum dried.

2.3. Induction of experimental diabetes in rats

This study was approved by the Animal Ethics Committee of Northwest A&F University. Female Sprague–Dawleys rats weighing 170–200 g were obtained from the Experimental Animal Center at the 4th Military Medical University, Xi'an, Shaanxi Province. Rats were allowed to acclimatize for 7 days in an environmentally controlled room at 22 °C with an alternating 12 h light/dark cycle and free access to normal laboratory food and water. After 1 week acclimation, rats in the challenge group were given high-fat diet consisting of 10% lard, 20% sucrose, 2.5% cholesterol, 1.0% cholate and 66.5% normal laboratory food. Rats in the control group

Table 1
Experiment groups and treatments.

Treatment groups	Type of animals	Formula of feed
Normal control group	Normal rats	Normal laboratory food
Diabetic control group	Diabetic rats	High-fat diet
Metformin control group	Diabetic rats	High-fat diet + metformin (70 mg/kg)
0.3% ASK group	Diabetic rats	High-fat diet + 0.3% ASK gum
0.9% ASK group	Diabetic rats	High-fat diet + 0.9% ASK gum
2.7% ASK group	Diabetic rats	High-fat diet + 2.7% ASK gum

containing 8 rats were continually fed only normal laboratory food. All rats could eat and drink *ad libitum*. After 6 weeks of the high-fat diet, rats were fasted for 12 h (free access to water) and each rat was injected with 30 mg/kg streptozotocin (STZ; Sigma, St. Louis, MO, U.S.A.) in 0.1 M acetate buffer, pH 4.5. Rats in the control group were injected with only the buffer solution. Three weeks after the injection, rats were fasted for 12 h (free access to water) and treated with D-glucose (2 g/kg, p.o.). The blood glucose levels at 0 min and 120 min after the glucose treatment were determined. Rats with blood glucose level ≥ 7.8 mmol/L at 0 min or ≥ 11.1 mmol/L at 120 min were considered to be diabetic.

2.4. Experimental groups and treatments

From the above high-fat-diet treated rats, 40 rats that have developed type 2 diabetes were chosen with the closest body weight. The animals were randomly divided into five treatment groups as described in Table 1 and each group contained eight animals. The rats in the previous control group were continually given normal laboratory food and served as the normal control; while diabetic rats received the high-fat diet containing different amounts of ASK gum or metformin in replacement of the normal laboratory food portion (66.5%), and the same amount of lard (10%), sucrose (20%), cholesterol (2.5%) and cholate (1.0%). Metformin is used as a positive control because of its well known anti-oxidative effects in the control of diabetes mellitus (Rasilainen, Jenni, Levo-nen, Otonkoski, & Lapatto, 2002). Each treatment lasted for eight weeks during which period rats had free access to food and water. For clarity, the experiment scheme is summarized in Fig. 1.

2.5. Sample collection

After above treatments, rats were fasted for 12 h, anaesthetized between 8:00 a.m. and 9:00 a.m. using ketamine (24 mg/kg body weight, intramuscular injection). Blood from the celiac artery was collected into dry test tube using syringes and allowed to coagulate at ambient temperature for 30 min. The serum was collected for further analysis after centrifugation at 200 g for 10 min. Livers and kidneys were immediately dissected out, washed in ice-cold saline solution to remove the blood, then kept in 10% formalin solution for the histopathological study. Each liver sample was sliced into pieces

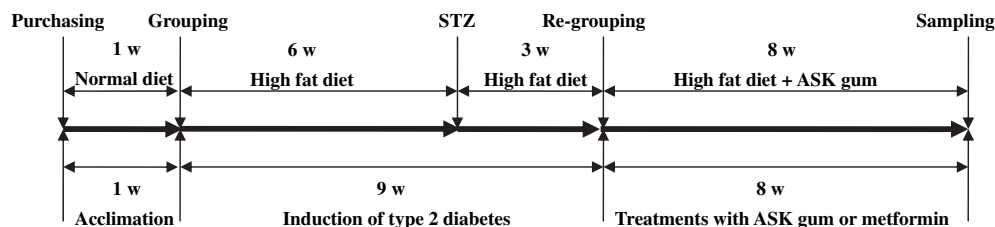


Fig. 1. Experiment scheme.

and homogenized in cold phosphatic buffer solution (pH 7.0) to give a 10% homogenate (w/v). The homogenates were centrifuged at 50 g for 10 min at 0 °C and the supernatants were taken for the hepatic biochemical estimation.

2.6. Bioassay of blood and liver

SOD, MDA and •OH in serum and liver were determined on a UV9100 spectrophotometer (Beijing Ruili Analytical Instrument Corp., Beijing, China) using commercial kits purchased from the Nanjing Jiancheng Bioengineering Institute (Nanjing, China). Briefly, the assay for SOD was based on its ability to inhibit the oxidation of oxymine by the xanthine–XO system (Oyanagui, 1984). The MDA assay was based on that MDA reacts with thibabituric acid (TBA) to form a product which has maximal absorption at 532 nm (Ohkawa, Ohishi, & Yagi, 1979). The assay for •OH was based on Fenton reaction (Ruiz, Leal, Martin, Martinez, & Lacort, 1995).

2.7. Histopathological study

Kidney samples were firstly fixed in 10% formal-saline for 48 h, and dehydrated by successively passing through a gradient of mixtures of ethyl alcohol and water. The dehydrated samples were then embedded in paraffin after rinsing with xylene. Kidney sections (5 µm thick) were prepared and stained with hematoxylin and eosin dye, before mounted in neutral deparaffinated xylene (DPX) medium for microscopic examination on a Motic BA 400 microscope with a Motic Advance 3.0 software.

2.8. Statistical analysis

The statistical significance of differences among the six groups was assessed using analysis of covariance (ANOVA) by SAS software.

3. Results

3.1. Yield of extraction and chemical composition of the ASK gum

The current extraction method yielded 36.6% of ASK gum based on dry ASK seed powder. The major chemical components of ASK gum are 62.0% total sugar, 23.1% uronic acids, 12.9% protein, and 9.5% ash on the dry weight bases. Our previous study reported that ASK polysaccharide contained a mixture of monosaccharides including glucose, galactose, mannose, xylose, arabinose and rhamnose (Hu et al., 2006).

3.2. Establishment of diabetic rat model

After STZ injection and 9 weeks high-fat diet feeding, all rats in the diabetic group developed type 2 diabetes; the blood glucose levels at 0 min and 120 min were significantly higher than that in the normal group ($P < 0.01$) as shown in Table 2. Compared to the rats in the normal group, the body weight of rats in the diabetic group were also significantly higher ($P < 0.01$), although the feed

Table 2

The evaluation of diabetic model after 9 week high-fat diet and STZ treatment.

Parameters	Normal rats ($n = 8$)	Diabetic rats ($n = 40$)
0 min blood glucose (mmol/L)	4.45 ± 0.60	20.14 ± 1.34a
120 min blood glucose (mmol/L)	5.30 ± 0.85	23.48 ± 1.75a
Body weight (g)	320 ± 12	395 ± 14a

Note: The letter “a” indicates $P < 0.01$ when compared with normal rats.

consumption in the former was higher than the latter (Table 3). This was due to the higher calorie of high-fat diet compared to the normal laboratory diet. It was found that the supplementation of ASK gum did not impact the feed consumption of rats during the treatment (Table 3).

3.3. Effect of ASK gum on the levels of SOD, MDA, and •OH in liver

Liver SOD activities of rats in the diabetic groups were significantly ($P < 0.01$) lower than those in the normal group (Fig. 2A). The liver SOD activity increased gradually as the ASK gum concentration in the diet increased. The liver SOD was significantly ($P < 0.01$) higher in the 2.7% ASK gum group compared to the diabetic control group. There was no significant difference ($P > 0.05$) between the 2.7% ASK gum group and the metformin control group.

Liver MDA levels of diabetic rats were significantly ($P < 0.01$) higher than that of normal rats (Fig. 2B). As the ASK gum content in the high-fat diet increased, there was a gradual decline in the liver MDA level. The liver MDA level in the 2.7% ASK gum treatment was significantly ($P < 0.01$) lower than the diabetic control group but similar to that in the metformin control group ($P > 0.05$).

Liver •OH level in the normal control treatment was significantly ($P < 0.01$) lower than that in all the diabetic groups (Fig. 2C). Liver •OH level in the 0.3% ASK gum treatment was similar to the diabetic control treatment ($P > 0.05$). In contrast, liver •OH level in the 0.9% and 2.7% ASK gum treatments were significantly lower compared to the diabetic control group ($P < 0.01$). There was no significant difference ($P > 0.05$) in the liver •OH levels between the 2.7% ASK gum treatment and the metformin treatment.

3.4. Effect of ASK gum on SOD, MDA and •OH in serum

Serum SOD activities of diabetic groups were significantly ($P < 0.01$) lower than that in the normal control group (Fig. 3A), however, the groups treated with ASK gum showed much higher SOD activities than the diabetic control group ($P < 0.05$). There was no significant difference ($P > 0.05$) in serum SOD activity among the three ASK gum groups and the metformin control group.

Serum MDA levels of diabetic groups were significantly ($P < 0.01$) higher than that in the normal group (Fig. 3B). The 0.3% ASK gum treatment group was similar to the diabetic control group

Table 3

Average daily feed consumption (gram per rat) during ASK treatment (from 9th week to 17th week).

Groups	Day 6	Day 12	Day 18	Day 24	Day 30	Day 36	Day 42	Day 48	Day 54
Normal control	24.20	25.23	26.17	27.10	27.89	28.51	29.24	29.85	30.28
Diabetic control	21.47	21.92	22.35	22.72	23.16	23.35	23.69	23.89	24.15
Metformin control	21.23	21.58	22.10	22.54	22.92	22.99	23.40	23.81	23.97
0.3% ASK gum	21.44	21.97	22.35	22.80	23.11	23.47	23.71	24.00	24.34
0.9% ASK gum	21.34	21.87	22.39	22.75	22.97	23.35	23.65	23.97	24.39
2.7% ASK gum	21.65	22.07	22.27	22.57	22.86	23.31	23.77	24.08	24.44

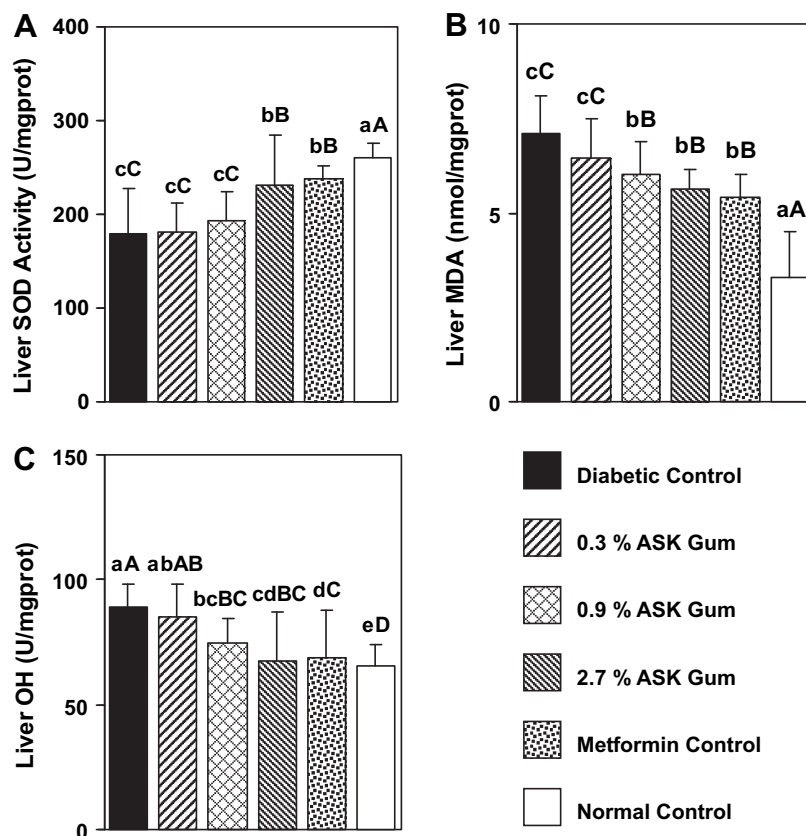


Fig. 2. Effect of ASK gum on: (A) liver SOD activity, (B) liver MDA level, and (C) liver $\cdot$ OH. Bars with different uppercase letters are significantly different at $P < 0.01$. Bars with different lowercase letters are significantly different at $P < 0.05$.

in serum MDA level ($P > 0.05$); however, serum MDA levels in the 0.9% and 2.7% ASK gum group were significantly lower ($P < 0.05$) than that in the diabetic control group. These results suggest that incorporation of 2.7% ASK gum into the diet provided the same protection effect ($P > 0.05$) as metformin did.

Rats in all diabetic groups had significantly ($P < 0.01$) higher serum $\cdot$ OH levels compared to the rats in the normal control group (Fig. 3C). As the amount of ASK gum in the diet increased, serum $\cdot$ OH levels in diabetic rats gradually decreased. The 2.7% ASK gum group had a significantly ($P < 0.05$) lower $\cdot$ OH level compared to the diabetic control group and a similar level to the metformin control group ($P > 0.05$).

3.5. Effect of ASK gum on renal histological structure of type 2 diabetic rats

Compared to the normal control group, renal tubules of rats in the diabetic groups demonstrated different degrees of oxidative damage (Fig. 4). Severe oxidative damage was observed for the rats in the diabetic control group as evidenced by greater amount of renal cells suffered from abnormal apoptosis: these dead cells fell off from renal tubules, leaving many large blank areas as shown in Fig. 4A; clear brush border was hardly seen for these renal tubules. In contrast, there was a significant reduction in the histological degeneration of renal tubules among the ASK gum groups. As the amount of ASK gum increased, the histological conditions also improved. Most of the renal tubules in the 2.7% ASK gum group could be easily recognized (Fig. 4D), and the appearance was very similar to that of the metformin control group (Fig. 4E).

4. Discussions

4.1. Some considerations regarding the animal model and the method of gum supplementation

STZ is an antibiotic produced by *Streptomyces achromogenes* (Rakieten, Rakieten, & Nadkarni, 1963). It is known that STZ is cytotoxic to β -cell and is a mild carcinogenic substance. It is commonly used in the establishment of experimental diabetes animals (Coskun, Kanter, Korkmaz, & Oter, 2008). It has been suggested that Sprague–Dawleys rats fed with high-fat food and injected with a low-dose of STZ could serve as an alternative animal model for simulating type 2 diabetes in humans (Mikio, Hideyuki, & Tsutomu, 2006; Reed et al., 2000; Srinivasan, Viswanad, Asrat, Kaul, & Ramarao, 2005; Zhang, Ye, & Li, 2003). These researchers used this model to test anti-diabetic and antioxidant agents for the treatment of type 2 diabetes. In the current study, we used a similar protocol to reproduce the type 2 diabetic rat model. The significant higher fasting and postprandial blood glucose levels in the treated rats compared to that in the control rats indicated that this high-fat diet and low-dose-STZ protocol successfully induced hyperglycemia in rats. We also observed that body weight of such treated rats were significantly higher than those of control rats. This suggested that the STZ-induced diabetic rats acquired type 2 diabetes rather than type 1 diabetes. This is because, unlike type 1 diabetes which causes body weight reduction, type 2 diabetes usually elevates body weight (Reed et al., 2000). In our previous study which also used this animal model, we found that, after STZ injection and 16 weeks high-fat diet feeding, the rats in diabetic control group demonstrated severe hyperglycemia, hyperlipemia and insulin resistance

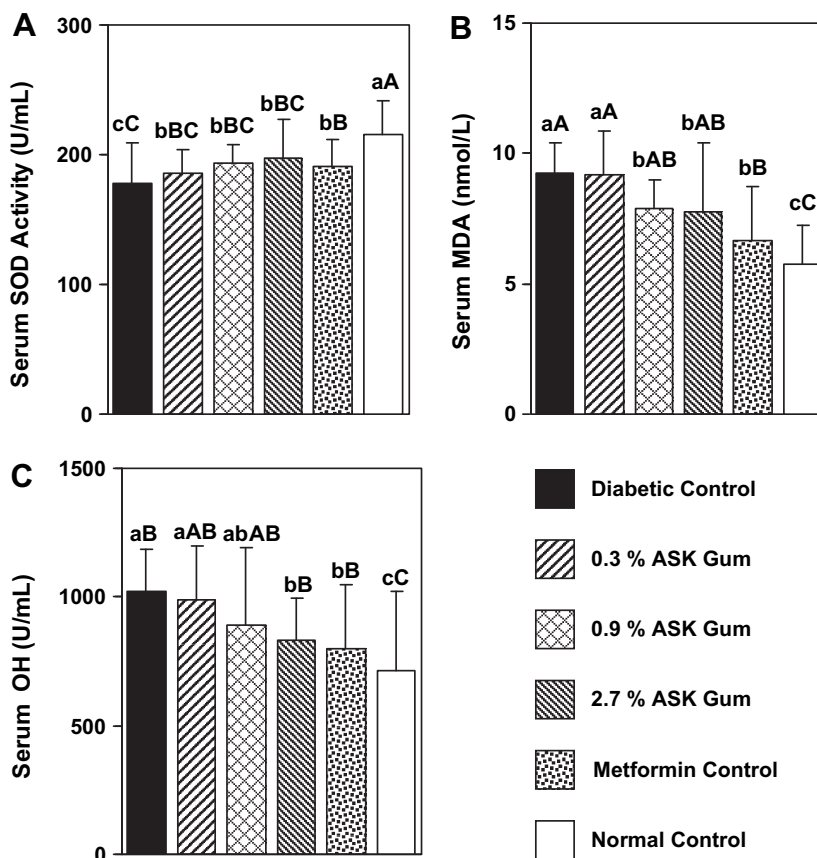


Fig. 3. Effect of ASK gum on: (A) Serum SOD activity, (B) Serum MDA level, and (C) Serum $\cdot\text{OH}$. Bars with different uppercase letters are significantly different at $P < 0.01$. Bars with different lowercase letters are significantly different at $P < 0.05$.

which were characteristics of type 2 diabetes (unpublished data). Therefore, we are confident that we have successfully induced type 2 diabetes mellitus in Sprague–Dawleys rats.

In the current study, ASK gum was administered to the rats through their food source rather than by injection or oral gavage, with an intention to closely mimic supplementation of ASK gum through human daily diet. The amount of ASK gum added to the test feed was similar to the level of ASK gum that is commonly added to food for improving food texture. We did not observe significant differences in feed consumption between the groups with or without ASK supplementation during the examination, indicating the amount of ASK used did not negatively impact the feed taste and/or preference. This implies incorporation of ASK gum into food may not only improve the texture and quality of the fortified food, but also could provide potential health benefits to the consumers.

4.2. Effect of ASK gum on lipid peroxidation in type 2 diabetic rats

Lipid peroxidation is a well recognized mechanism by which ROS cause oxidative damage to animal tissues. SOD is one of the major scavenging enzymes that remove ROS *in vivo*. A decrease in the activity of SOD can lead to an excess availability of superoxide anion $\text{O}_2^{\cdot-}$ and hydrogen peroxide in biological systems, which in turn generate hydroxyl radicals, resulting in initiation and propagation of lipid peroxidation (Kumuhekar & Katyane, 1992). In this study, diabetic rats showed low serum and liver SOD levels, which was in accordance with previous reports on animals (Feillet-Coudray et al., 1999) and human patients (Vucic, Gavella, Bozиков,

Ashcroft, & Rocic, 1997). ASK gum demonstrated antioxidant effects by significantly elevating serum and liver SOD levels in a dose-dependent manner. Accompanied by the elevation in SOD activity, we also observed a significant decrease in $\cdot\text{OH}$ level by ASK gum, in both serum and liver, in diabetic groups. This confirmed that ASK gum could help to scavenge $\text{O}_2^{\cdot-}$, inhibit the generation of $\cdot\text{OH}$, and reduce lipid peroxidation. This action could involve mechanisms related to antioxidant effect of ASK gum on type 2 diabetes.

The measurement of the MDA provides a convenient index of lipid peroxidation (Devasena, Lalitha, & Padma, 2001; Nielsen, Mikkelsen, Nielsen, Andersen, & Grandjean, 1997). MDA is formed as an end product of lipid peroxidation. The intermolecular crosslink of collagen through MDA leads to its stabilization and allows further glycation. This in turn increases the potential of glycated collagen to initiate further lipid peroxidation and releasing more MDA. Thus, MDA could be a link between glycation and further lipid peroxidation (Bhatia et al., 2003; Slatter, Bolton, & Bailey, 2000). Our study found that a high-fat diet elevated MDA levels in both serum and liver of diabetic rats; inclusion of ASK gum in this diet attenuated the rises in serum and liver MDA in a dose-dependent manner. This further confirmed that ASK gum had the ability to suppress lipid peroxidation in type 2 diabetes.

Due to the limitation on the size of the experiment, this study did not include the treatment of diabetic rats under normal diet with ASK gum. Further studies to evaluate the antioxidant effects of ASK gum in type 2 diabetes rats under a normal non-high-fat diet will be conducted in a separate study.

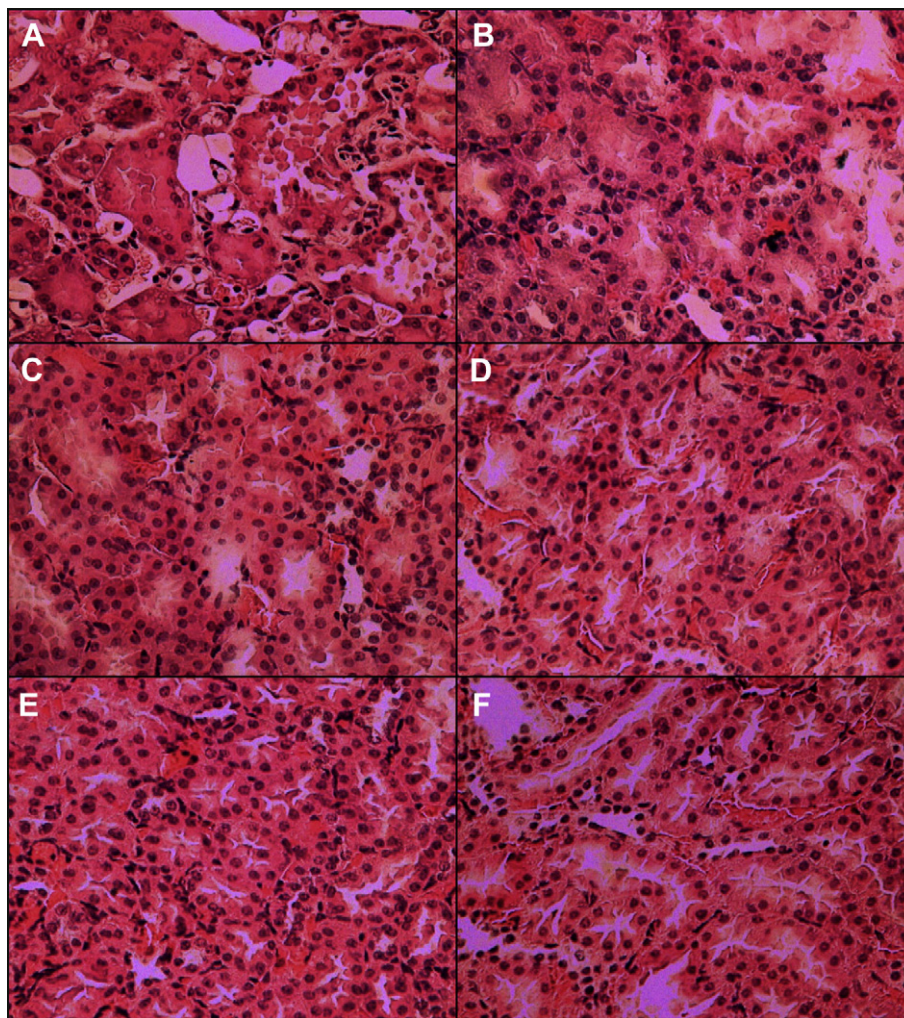


Fig. 4. Effect of ASK gum on oxidative damage of renal tubule in different groups of rats: (A) Diabetic Control, (B) 0.3% ASK Gum, (C) 0.9% ASK Gum, (D) 2.7% ASK Gum, (E) Metformin Control and (F) Normal Control. Original magnification, 400 $\times$.

4.3. Effect of ASK gum on oxidative damage of tissue in type 2 diabetes

ROS can react with cellular components, especially membrane lipids, producing cell damage (Bhatia et al., 2003). Diabetes mellitus is associated with generation of ROS leading to oxidative damage particularly in liver and kidney (Mohamed, Bierhaus, & Schiekofer, 1999). Our previous study observed severe fatty tissue degradation in STZ-induced type 2 diabetic rats and ASK gum alleviated the hepatic fatty degradation significantly (unpublished data). In the current study, ASK provided protection to the kidney of STZ-induced type 2 diabetic rats. All the results pointed to the fact that ASK had antioxidant effect on type 2 diabetic rats and could reduce oxidative damage in their tissues.

5. Conclusions

In summary, our results provide evidence that incorporation of ASK gum into a high-fat diet can alleviate oxidative stress of high-fat low-dose-STZ-induced type 2 diabetic rats. ASK gum can reduce the tissue oxidative damage induced by ROS in type 2 diabetes. Its antioxidant mechanism may be associated with the functions of ROS-scavenging enzymes, such as SOD.

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